



Investigation of the Potential of Human Urine as an Organic Corrosion Inhibitor for Cast Iron in Sodium Chloride Media

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Abstract

Gray cast iron is widely used in engineering because of its compressive strength, vibration damping, machinability, formability, and cost effectiveness, but it remains susceptible to corrosion. This study investigated aqueous ammonia derived from human urine as an organic corrosion inhibitor for cast iron in sodium chloride solution. Linear Polarisation Resistance was used to determine corrosion rates, while Scanning Electron Microscopy and Energy-Dispersive X-ray Spectroscopy were used to examine the corroded surfaces. Samples containing 5–15 cm³ of inhibitor exhibited lower corrosion rates than the uninhibited control, which recorded 0.264 mm/year. Sample A1, containing 5 cm³ of inhibitor, achieved the lowest corrosion rate of 0.166 mm/year and an inhibition efficiency of 37.05%. Samples containing 20–25 cm³ produced higher corrosion rates than the control; Sample E1 recorded the highest rate at 0.348 mm/year. The microscopy results were consistent with the electrochemical measurements, showing fewer corrosion products at concentrations of 5–15 cm³ and more corrosion products at 20–25 cm³. The results demonstrate limited corrosion inhibition under the evaluated conditions and show that performance depends strongly on inhibitor concentration. Keywords: cast iron, aqueous ammonia, organic inhibitor, corrosion, Linear Polarisation Resistance.

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1. Introduction

1.1 Background Study

Cast iron is an alloy of iron that is made by the reduction of iron ore in a blast furnace and contains more than 2% of carbon, silicon, manganese, and traces of Sulphur and phosphorus. It is hard, a good conductor of heat, affordable. Cast iron also possesses qualities like: high compressive strength, high vibrational damping, high machinability, formidability. Because of some of these properties it possesses various applications; for instance because of Cast iron's properties like its: high compressive strength, formidability, low cost, it is used in internal combustion engine's cylinder blocks, gear box cases, manifolds, flywheels, disk brake rotors.

Properties like non stickiness, heat conduction and its ability to distribute heat evenly following the development of maillard's reaction, a reaction between amino acids and reducing sugars which gives brown food their attractive taste makes cast iron to be used in making cooking ware (Millard, 1912; Stefanescu, 2018). Its hardness and low cost makes Cast iron to be used in making pipes, fences, stairs, lamp posts, farm equipment, processing equipment.

Cast iron's castability property is as a result of possessing the combination of high carbon content and Sulphur. As a result of various heat treatments and processing techniques, there are various types of cast iron namely: white iron, gray iron, ductile iron, malleable iron and compacted graphite iron.

With properties like strength, good castability, machinability, easy sterilization. Cast iron is being used in medical applications. For instance some radiotherapy molds, mortar and pestle for grinding and mixing pharmaceutical ingredients. Because it's high damping ability cast iron can be used as a bed, base and frames for machines. For the purpose of this project, I would be using Gray Cast Iron which is commonly used in the railway rails, construction of marine harbor and port components such as bollards, cleats and other mooring elements due to its strength and resistance to corrosion in seawater, ship rudders because of its damping capacity, ship propeller and propeller housing because of its strength and resistance to corrosion, marine deck cleats and marine deck fairleads because of its strength and durability in sea water. Properties like good castability, low cost (20-40% less than steel), flexibility makes gray cast iron to be chosen in various industrial applications (Hammood and Lieth, 2012; Kadhim, 2017). Gray Cast Iron also possess properties such as 100% recyclability, good resistance to friction and its resistance to compressive stresses 3-4 times of resistance to tensile stresses (Rajput, 2007; Black and Kosher, 2008; Kadhim, 2017).

However a major problem facing the use of cast iron in engineering application is corrosion. Corrosion is the harmful attack of a metal by electrochemical or chemical reaction with its environment (Anyanwu *et al.*, 2018). Corrosion reduces the mechanical properties of this metal thereby causing problems in structures built with this metal (Anyanwu *et al.*, 2021). For instance, cast iron used in railway rails built across sea water corrodes which lead to the destruction of

the railway rails (Guarneros *et al.*, 2021). Lack of conservation and restoration treatment on metal surfaces leads to the degradation of the metal which as a result lead to the discovery and invention of methods to prevent corrosion in cast iron which includes galvanization, painting and greasing, electroplating, inhibitors (Vázquez *et al.*, 2021). Galvanization is done by dipping iron in molten zinc. This zinc layer coating is used to protect the inner part of the iron from corrosion. Electroplating is a process that involves use of electric current to cover the thin layer of metal over the top of another less expensive metal. Painting and greasing involves the use of paint or grease to create a layer on the metal which prevents the metals from direct attack from the environment. Corrosion inhibitors are substances which, when added in small concentration to a material effectively reduces the corrosion rate of a material exposed to that environment (Patni *et al.*, 2013). Inhibitors can be natural or synthetic inhibitors. Synthetic inhibitors are produced with chemicals so as to provide high success rate in various corrosive environments. Examples include amines salts, phosphates, nitrites (Al-Amiery *et al.*, 2023).

Natural inhibitors which are derived from plants, animals are ecofriendly, biodegradable and possess low toxicity (Al-Amiery *et al.*, 2023). Aqueous ammonia which can be extracted from human urine which is eco-friendly, readily available, inexpensive has being discovered to be a corrosion inhibitor.

Shuaib and Babata (2004) investigated the corrosion inhibition of mild steel in ammonia solution. Weight loss technique was used to evaluate the efficiency of ammonia solution as a corrosion inhibitor for mild steel. It was discovered that ammonia solution is a good inhibitor for corrosion in mild steel.

Verma *et al.*, (2020) investigated the use of ammonia derivatives as corrosion inhibitors. It was concluded that aliphatic and aromatic amines, amino acids, hydrazine and phenylhydrazine and their derivatives are most frequently used ammonia derivatives as corrosion inhibitors.

1.2 Problem Statement

Gray cast iron employed for the construction of railway rails along coastal axis in Nigeria deteriorates in properties over time as a result of corrosion. This is connected to the effect of too much chloride ions in the soil and water body where they are installed (Guarneros *et al.*, 2021). The cost of salvaging corrosion of gray cast iron in railway rails is usually high and runs in to several millions of naira. There is need to formulate green inhibitors that can help reduce the effects of corrosion products in gray cast iron. Green inhibitors are cheap, easily sourced and harnessed. They are also eco-friendly and highly efficient when it comes to curbing corrosion.

1.3 Aim and Objectives

The aim of this research is to study the potentials of human urine as an organic inhibitor for cast iron corrosion protection in sodium chloride solution.

The specific objectives of this research are to:

- I) Formulation of ammonia based inhibitor from human urine.

- II) Determination of the corrosion rate of cast iron inhibited by ammonia solution in 3.5% of sodium chloride by Linear Polarization Resistance method.
- III) Determining the morphology of the samples via Scanning Electron Microscopy (SEM).

2. Literature Review

This chapter focuses on literature survey relevant to the potential of aqueous ammonia derived from human urine as a corrosion inhibitor for corrosion for cast iron in sodium chloride solution.

2.1 Review of previous researches in literature

Various inhibitors have been discovered and invented to curb corrosion of cast iron. Inhibitors can be synthetic or natural. Synthetic inhibitors are produced with chemicals so as to provide high success rate in various corrosive environments. Examples include amines salts, phosphates, nitrites (Al-Amiery *et al.*, 2023).

Natural inhibitors on the other hand are made from the plants, animals, minerals. They are eco-friendly, renewable, readily available, and safe for human exposure which makes to be suitable for application in food and pharmaceutical industries (Al-Amiery *et al.*, 2023).

Babata and Shuaib (2004) investigated the corrosion inhibition of mild steel in ammonia solution. Weight loss technique was used to evaluate the efficiency of ammonia solution as a corrosion inhibitor for mild steel. It was discovered that ammonia solution is a good inhibitor for corrosion in mild steel.

Using Schiff Base Derivatives (2-acetylpyridine thiosemicarbazone (I1), 2-acetylpyridine-(4-methylthiosemicarbazone) (I2), and 2-acetylpyridine-(4-phenylthiosemicarbazone) (I3) as corrosion inhibitors for cast iron in acid, base and neutral media. The three Schiff base derivatives namely; 2-acetylpyridine thiosemicarbazone (I1), 2-acetylpyridine-(4-methylthiosemicarbazone) (I2), and 2-acetylpyridine-(4-phenylthiosemicarbazone) (I3) were tested against corrosion of cast iron in aqueous solutions of HCl, NaOH, NH₄Cl, and NaCl and their efficiencies were measured by mass loss method and electrochemical measurements. The efficiency of inhibition decreases with prolonged immersion time and at low temperatures and is directly correlated with the concentration of the inhibitor (Rajeswari *et al.*, 2013).

Rajeswari *et al.*, (2014) investigated the effects of Corrosion inhibition of Eleusine aegyptiaca and Croton rottleri leaf extracts on cast iron surface in 1 M HCL medium. The efficiency of the Eleusine aegyptiaca and Croton rottleri leaf extracts as corrosion inhibitors on cast iron surface was investigated by weight loss, Tafel polarization and electrochemical impedance spectroscopy. The outcomes acquired from the weight loss and electrochemical techniques showed that the restraint proficiency expanded with inhibitor fixations.

Sheriff *et al.*, (2015) performed research on the corrosion inhibition of cast iron in the Arabian Gulf seawater by two different ionic liquids. The two ionic liquids that were used were synthetic inhibitors namely; 1-ethyl-3-methylimidazolium chloride ([EMIm]Cl) and 1-butyl-1-methylpyrrolidinium chloride ([Py1,4]Cl). The inhibiting influences of the ionic liquids was scrutinized by weight loss or gravimetric method, open circuit potential electrochemical impedance spectroscopy, and cyclic potentiodynamic polarization. The results showed that

(EMIm)Cl shows more inhibition efficiency at a short immersion time, for the examined ionic liquids concentrations compared to (Py1,4)Cl. However, (Py1,4)Cl shows more efficiency upon increasing the immersion time indicating the persistence of the inhibiting influence. The adsorption of the cations of the ionic liquids onto the surface of cast iron forming a corrosion barrier makes the ionic liquids used to inhibit corrosion.

Using Cantaloupe (*Cucumis melo*) as a green inhibitor of corrosion for Cast Iron in Hydrochloric (HCL) acid. The efficiency of this inhibitor was investigated by using Hydrogen Evolution Measurements (HEM) and Mass Loss Measurement (MLM). Cantaloupe extracts inhibited corrosion of Cast Iron in 1.0 M of HCl solution. As the concentration of the cantaloupe extracts was increased, the inhibiting efficiency of the green inhibitor also increased (Al-Ahmadi *et al.*, 2015).

Vázquez *et al* (2021) investigated the effects of corrosion inhibition with xanthines in gray cast iron protection with Paraloid B-72 in a saline medium. Electrochemical Impedance Spectroscopy was used to investigate the efficiency of caffeine and theophylline concentrations which are xanthines as corrosion inhibitors for grey cast iron, in combination with a coating (Paraloid B-72) in a saline medium. The two inhibitors showed promising results as far as corrosion inhibiting is concerned. In high concentrations, caffeine showed inhibition against corrosion, while at low concentrations, Theophylline exhibited corrosion inhibition. Resulting to a synergic effect with the Paraloid B-72 by the two different xanthine concentrations.

2.2 Literature with respect to corrosion

Corrosion

Corrosion is the harmful attack of a metal by electrochemical or chemical reaction with its environment (Revie, 2018). Corrosion is the point at which a refined metal is normally switched over completely to a more steady structure, for example; its oxide, hydroxide or sulfide which results to the weakening of the material. Corrosion damage metals by converting them into oxide or other corrosion production (McCafferty E., 2010). Factors affecting corrosion includes: increase in temperature, presence of impurities such as salts, presence of moisture, exposure of the metals to air conditioning gases (SO₂, CO₂).

Types of Corrosion

Various types of corrosion are caused by the environment that the metals are exposed to. They include:

Uniform Corrosion

In Uniform Corrosion, a uniform layer of rust is formed on the outer layer of the metal and spread over the whole surface of the metal. This kind of corrosion is generally seen on metals that are not safeguarded by surface covering. Metals like zinc and Lead are commonly affected by uniform corrosion (Vedantu, 2024). Below is an image of uniform corrosion;



Figure 2.1: Uniform Corrosion (Vedantu, 2024)

Fretting Corrosion

This type of corrosion occurs at the point of contact of materials that are joined together. This types of corrosion is commonly found on bolt and riveted joints, clamped surfaces. This usually appears when the contact area is subjected to slips and variations (Vedantu, 2024). Below is an image of fretting corrosion;



Figure 2.2: Fretting Corrosion (Vedantu, 2024)

Erosive Corrosion

The mechanical abrasion due to the relative movement between the metal surface and corrosive liquids is referred to as Erosive corrosion. This type of corrosion commonly occur in metals carrying fluids in them. In this, the surface of metals gets damaged gradually by the abrasion of fast-moving liquid, and cavities are also formed (Vedantu, 2024). Below is an image of erosive corrosion;



Figure 2.3: Erosive Corrosion (Vedantu, 2024)

Intergranular Corrosion

This corrosion appears when there is a noticeable difference in reactivity against impurities between grains and grain boundaries. The difference in reactivity could be as a result of defective welding, heat treatment etc. Intergranular corrosion is the corrosion that appears on the granular region (Vedantu, 2024). Below is an image of intergranular corrosion;

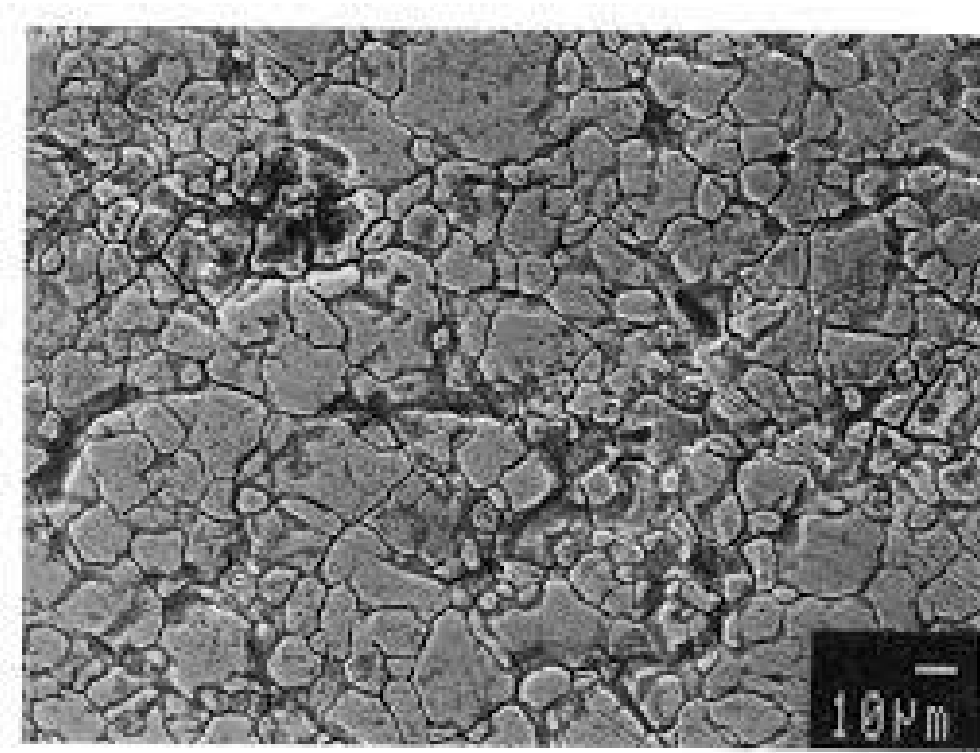


Figure 2.4: Intergranular Corrosion (Vedantu, 2024)

Corrosion Fatigue

If the failure of material happens in a corrosive environment, this occurrence is called Corrosion fatigue (Vedantu, 2024). It occurs when corrosion of a material takes places as a result of the material being exposed to cyclic loads. Below is an image of corrosion fatigue;



Figure 2.5: Corrosion fatigue (Vedantu, 2024)

Crevice Corrosion

This is the type of corrosion that is caused by small volumes of stagnant solutions in holes, gasket surfaces, lap joint, crevices under bolts and rivet heads. Below is an image of crevice corrosion;



Figure 2.6: Crevice corrosion (Vedantu, 2024)

Pitting Corrosion

This is type of corrosion where localized small holes/cavities are formed in a material. They are difficult to detect, prevent, manage or correct. Below is an image of pitting corrosion;

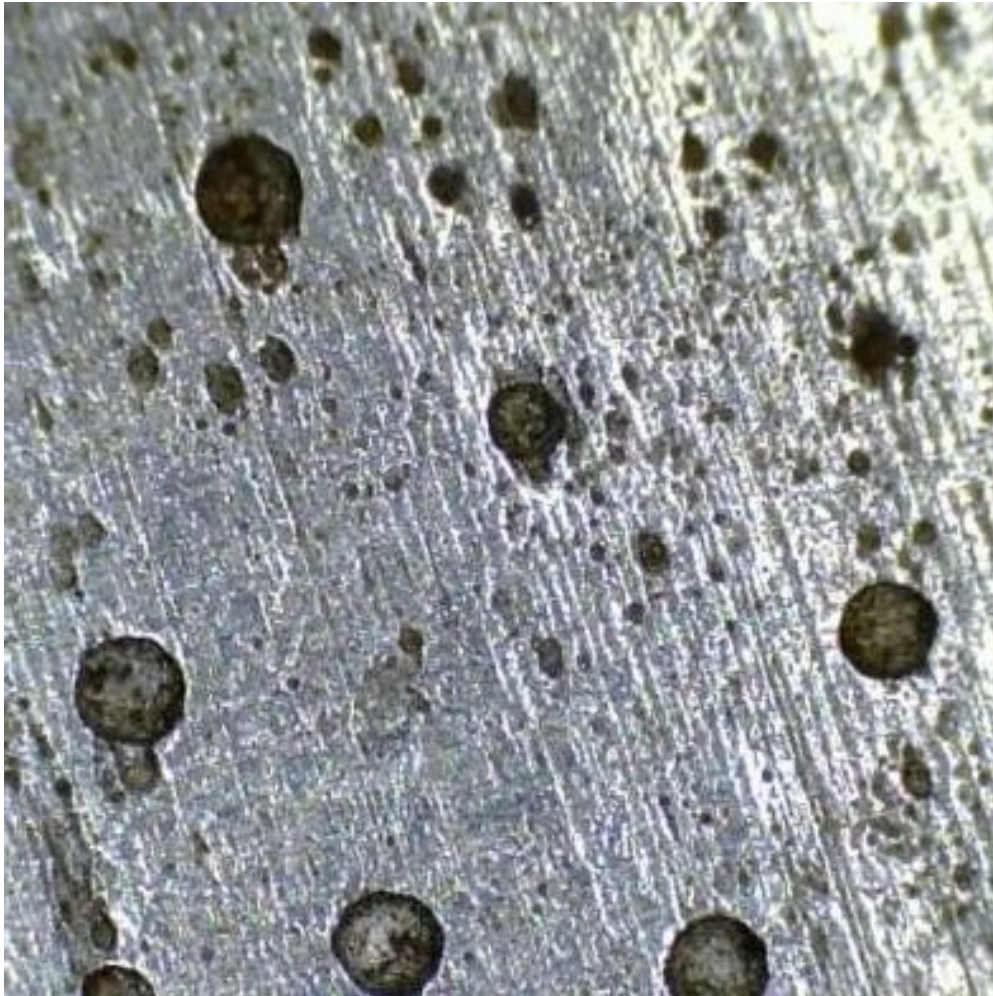


Figure 2.7: Pitting corrosion (Vedantu, 2024)

Corrosion prevention

Corrosion can't be eradicated totally but can be reduced. There are various ways to prevent corrosion, this include:

Electroplating

This is a process that uses electric current to cover the thin layer of metal over the top of another less expensive metal. It is usually done to provide finishing to the cheaper metals.

Galvanization

Galvanization is done by dipping iron in molten zinc. This zinc layer coating is used to protect the inner part of the iron from corrosion.

Painting and greasing

The use of paints and grease creates a layer on the material which helps to fight against corrosion.

Selection of optimum materials

Materials like Stainless steels and aluminium are highly corrosion resistant. The applications of this material helps to combat corrosion.

Use of inhibitors

Inhibitors are substance which when applied to materials reduce or prevents the rate of corrosion.

There are other ways to protect the metal from corrosion such as alloying, cathodic treatment protection, anodic protection and recently been using the laser for this purpose by surface of metal is considered as the way to improve the properties of metals like roughness, hardness, the resistance of corrosion(Kadhim *et al.*, 2021).

The use of inhibitors is one of cheapest and effective means of preventing corrosion. Corrosion inhibitors are of considerable practical importance, as they are extensively used in reducing metallic waste during production and in reducing the risk of material failure, both of which can lead to shut down of industrial processes, which in turn leads to additional costs (Kadhim *et al.*, 2021).

INHIBITORS

Corrosion inhibitors are substances which, when added in small concentration to a material effectively reduces the corrosion rate of a material exposed to that environment (Patni *et al.*, 2013).

Corrosion inhibitor is a chemical substance which when applied to a corrosion system at a suitable concentration decreases the corrosion rate, without significantly changing the concentration of any corrosive agent (Monticelli, 2018).

They are generally effective in small concentrations. A corrosion inhibitor is a substance that slows down the rate at which a metal in the environment it is exposed to corrodes (Umoren *et al.*, 2022).

Corrosion inhibitors are one of the many different methods that conservation-restoration professionals have available to protect and prolong the life of metallic cultural heritage (Cano and Lafuente, 2013).

Corrosion inhibitors are chemical substances that, when added in small amounts to the environment in which a metal would corrode, will reduce, slow down or prevent corrosion of the metal (Damborenea and Arena, 2014). Since the 1940s, inhibitors have reduced corrosion rate without significantly changing the concentration of any corrosive agent. Organic compounds containing nitrogen atoms, such as imidazolines, amides, amidoamines, amines, and their salts have been successfully used as corrosion inhibitors in oil and gas pipelines carrying hydrocarbons containing CO₂ (Ghalib, 2023). Corrosion inhibitors may be classified as cathodic, anodic, or mixed, depending on whether their influence is mainly in retarding the cathodic or anodic re-

action of the corrosion process or both of them. As a result, they cause a shift of the corrosion potential of the inhibited metal toward respectively either to the positive or the negative directions or they substantially leave the metal corrosion potential more or less unchanged (Monticelli, 2018).

The various ways inhibitors reduces corrosion of metal include: Passivation, poisoning, precipitation, and adsorption on the surface of the metal (Umoren and saji, 2022). Oil and gas exploration and production, petroleum refining, chemical manufacture, heavy manufacturing, water treatment, and the product additive industries are all key users of corrosion inhibitors (Umoren and saji, 2022).

Corrosion inhibitors aid in extending the life of equipment, minimize system failures and shutdowns, eliminate product contamination, limit heat loss, and maintain the appealing appearance of a structure (Umoren and saji, 2022).

3. Materials and Methods

The experimental methods and procedures were adapted from established corrosion-testing literature.

3.1 Materials

Gray Cast Iron

It is the most important material in this study. This is the material on which the inhibitor performance was evaluated in the sodium chloride solution.

Human Urine

The urine provided the urea-containing feedstock used to produce the aqueous-ammonia inhibitor through reaction with sodium hydroxide and collection in distilled water. The urine sample was supplied by the researcher.

Hot plate and beakers

The beakers are used to contain the urine and the hot plate is used to heat the urine to a concentrated form of urea by converting the water present in the urine to vapor. This was also used to boil the sodium hydroxide solution to temperature where the solution reacts with urea to produce ammonia.

Stir bar

This is used to break down the bubbles in the urine as it starts boiling. This is very important because as the urine starts to get more concentrated it produces more foam which is hard to break down with a stir bar.

Sodium hydroxide solution

This is solution which reacts with urea at high temperature to form ammonia gas.

Latex tube

This is channel through which the ammonia gas goes into the Erlenmeyer flask.

Distilled water

This is used to ensure the ammonia gas gets converted to aqueous ammonia unlike in tap water where the ammonia is likely to react with carbonates to ammonium carbonates which isn't soluble and will precipitate out.

Erlenmeyer flask

This is a container where the distilled water and the ammonia gas react and form aqueous ammonia.

Linear Polarization Meter

This is a machine which measures linear polarization resistance to determine corrosion rate.

Emery paper

An abrasive paper covered with abrasive particles which is used for sanding. This is for smoothening the surface of the cast iron.

Ethanol

Ethanol found in alcohol is used for cleaning surface of the cast iron so the residues after using the emery paper will be removed.

Acetone

It is a colorless liquid with a distinct smell and taste. It is used to remove the grease on the washed cast iron and makes sure there no more external material on the cast iron.

Sodium Chloride Solution

This is the environment the efficiency of the green inhibitor used on the cast iron is evaluated in.

Scanning Electron microscope

This is a microscope that produces images of the sample (cast iron) by scanning the surface with a focused beam of electrons. It creates images which contains data about the surface geology and the structure of the metal.

3.2 Methodology

Formulation of ammonia based inhibitor from human urine

The process was started by filling a beaker with urine then boiling to a temperature of 100°C. This process was then repeated until the urea becomes concentrated, the urea was then decanted from the minerals in the urine. A concentrated sodium hydroxide solution was prepared in an Erlenmeyer flask and boiled on a hot plate to a temperature between 60-100°C which is the temperature where the reaction between the urea in urine and the sodium hydroxide solution proceeds. The concentrated urine was poured in the sodium hydroxide solution and their reaction rapidly evolves ammonia gas, this ammonia was then passed through a latex tube into a second Erlenmeyer flask containing distilled water under continuous stirring. The reaction between the

ammonia gas and the distilled water forms aqueous ammonia (Dbx labs, 2021). Table 3.1 shows the concentrations of the ammonia based inhibitor that was used in 3.5% NaCl media:

Table 3.1: Concentration of inhibitors

Aqueous ammonia concentration (cm ³)	Sample ID
5	A1
10	B1
15	C1
20	D1
25	E1

2. Determination of the corrosion rate of the gray cast iron in 3.5% NaCl solution using the ammonia based inhibitor

1. Preparation of samples

The Gray Cast Iron sheet was cut into different coupons, each with a dimension of 1 cm × 1 cm. The coupon was cleaned, then polished with emery paper. The coupons were then washed with distilled water then each coupon was washed with ethanol for degreasing, rinsed with acetone then allowed to air dry and the samples were then stored in desiccators.

Determination of corrosion rate

Each gray cast-iron coupon was immersed in 3.5% of sodium chloride solution and aqueous ammonia solution of different concentrations of 0cm³, 5cm³, 10cm³, 15cm³, 20cm³, 25cm³. The corrosion rate was investigated with Linear Polarization Resistance Method. Polarization measurement was conducted using a potentiostat/galvanostat. The reference electrode that was used is silver/silver chloride (Ag/AgCl). The counter electrode was platinum and the working electrode is the 1 cm × 1 cm gray cast iron specimen. Parameters like corrosion current density (*i*_{corr}), corrosion potential (*E*_{corr}), anodic Tafel slope and cathodic Tafel slopes were derived from the potentiodynamic curves (Abechi et al, 2020).

The formula used to calculate inhibition efficiency is:

$$\%IE = \frac{I_1 - I_2}{I_1} \times 100, \quad (3.1)$$

where *I*₁ and *I*₂ are the current densities in the absence and presence, respectively, of a predetermined concentration of inhibitor (Abechi et al., 2020).

Adsorption isotherm

The mechanism of corrosion protection may be explained on the basis of adsorption behavior. Adsorption isotherms provide important clues to metal-inhibitor interactions (Loto, Loto and Fedotova, 2014). The various methods for testing adsorption isotherm that was used for this research are:

Langmuir adsorption isotherm:

The adsorption value K indicates the adsorption strength the higher the value, the higher the adsorption strength (Ituen *et al.*, 2017; Gusti *et al.*, 2023).

The Langmuir adsorption isotherm is given by:

$$\frac{C}{\theta} = \frac{1}{K} + C. \quad (3.2)$$

θ is degree of surface coverage, K is a value that describes the adsorption strength, C is the concentration of the inhibitor (Ituen *et al.*, 2017; Gusti *et al.*, 2023).

Frumkin adsorption isotherm:

In Frumkin adsorption, α which has a positive value indicates the attraction behavior of the inhibitor on the metal surface. A negative α value indicates inhibit metal surface (Ituen *et al.*, 2017; Gusti *et al.*, 2023).

The Frumkin adsorption isotherm is given by:

$$\log\left(\frac{\theta}{1-\theta}\right) = \log K + \frac{2\alpha\theta}{2.303}. \quad (3.3)$$

θ is degree of surface coverage, α is a parameter characterizing the interaction between the adsorbed particles, K is a value that describes the adsorption strength (Ituen *et al.*, 2017; Gusti *et al.*, 2023).

Freundlich adsorption isotherm:

When $0 < \frac{1}{n} < 1$ adsorption is believed to be easy and when $\frac{1}{n}$ is $>$ or $= 1$ the adsorption is believed to be difficult (Ituen *et al.*, 2017; Gusti *et al.*, 2023).

The Freundlich adsorption isotherm is given by:

$$\log \theta = \log K + \frac{1}{n} \log C. \quad (3.4)$$

θ is degree of surface coverage, K is a value that describes the adsorption strength, C is the concentration of the inhibitor (Ituen *et al.*, 2017; Gusti *et al.*, 2023).

Gibbs free energy

The expression for Gibbs free energy change of adsorption presented in Equation (4) was used to investigate the feasibility and nature of the adsorption

$$\Delta G_{\text{ads}} = -RT \ln(55.5K). \quad (3.5)$$

K is the adsorption isotherm constant and 55.5 is molar concentration of water in solution. R is universal gas constant and T is temperature in kelvin (Akinbulumo *et al.*, 2020). Negative Gibbs free energy value means that adsorption of the inhibitor on the metal surface is sponta-

neous, feasible and occurred according to the mechanism of physical adsorption while positive Gibbs free energy value means that the adsorption of the inhibitor on the metal surface is non-spontaneous and non-feasible.

Morphology of the sample

The morphology of the sample was examined by a Scanning Electron Microscope (SEM) with a magnification of $4,000\times$. The Energy Dispersive Spectroscopy in SEM was used to determine the chemical composition of the gray cast iron as-received. X-ray Diffraction (XRD) was also used to determine the phase evolution present in the gray cast iron as-received.

4. Results and Discussion

4.1 Results for Energy Dispersive Spectroscopy and X-Ray Diffraction

The chemical composition of the as-received gray cast iron can be seen in figure 4.1 and the phase evolution present of the as-received gray cast iron can be seen in figure 4.2.

From figure 4.1 below, it can be seen that the carbon content (2.2%) is within the range of conventional gray cast iron which lies between (2-5%).

This result is further confirmed from the XRD shown in figure 4.2 below, it can be seen that peak for Iron (II) oxide is the highest which complies with that of conventional gray cast iron.

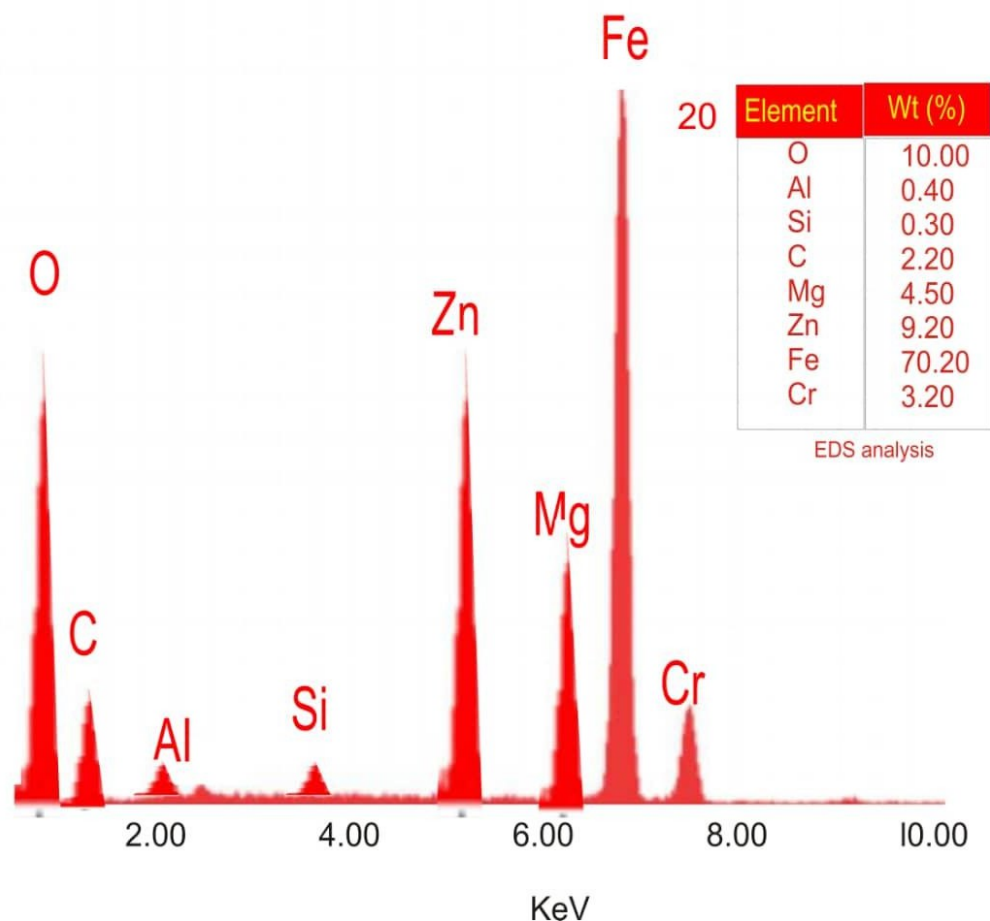


Figure 4.1: Chemical composition for gray cast iron

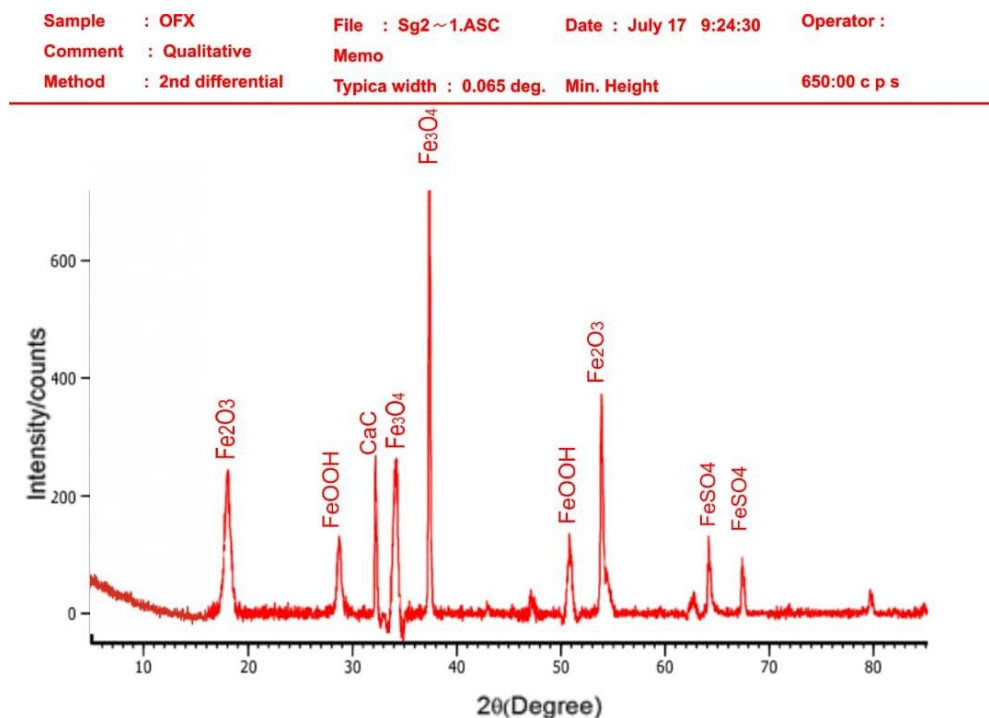


Figure 4.2: X-ray Diffraction of the gray cast iron

4.2 Corrosion Data of the Samples

Corrosion rate data of the samples

The corrosion-rate results of the samples used in this research are presented in table 4.1

Sample ID	Aqueous ammonia	
	Concentration (cm ³)	Corrosion Rate (mm/year)
Control 1	0	0.264
A1	5	0.166
B1	10	0.177
C1	15	0.235
D1	20	0.292
E1	25	0.348

Table 4.1 Corrosion rate results of the samples

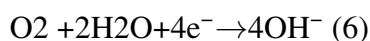
From table 4.1, it can be seen that the corrosion rate reduced when the concentration of the aqueous ammonia was between (5- 15cm³) just as Shuaib-Babata (2004) reported. It could also be seen that the corrosion rate increased when the concentration of the aqueous ammonia was increased (20-25cm³) which led to the increase of the concentration of ammonium chloride which increases the rate of corrosion just as Sun, A. and Fan, D., (2010) reported.

The results also show from table 4.1 that the corrosion rate is the least in sample A1 with a corrosion rate of 0.166mm/year. This means the most effective and suitable concentration of

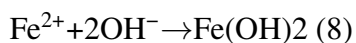
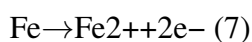
inhibitor identified in this study is that of sample A1. The corrosion rate was highest in sample E1 with a corrosion rate of 0.348mm/year. This means sample E1 is the least effective condition.

The corrosion rate reduced because the inhibitor formed a passivating layer on the surface of the cast iron which prevented the chloride ion from attacking the cast iron when concentration of the inhibitor was low which was in samples A1, B1, C1 (Shuaib-Babata, 2004). The corrosion rate increased when the concentration of the aqueous ammonia was increased in samples D1 and E1 which led to the increase of the concentration of ammonium chloride which increases the rate of corrosion (Sun, A. and Fan, D., 2010).

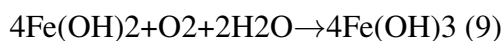
The corrosion rate mechanism involved is accompanied with the cathodic reaction shown in equation (6)



The anodic reaction involves the dissolution of the samples in order to balance the charge shown in equation (7) with the overall reactions in equation (8)



This ferrous hydroxide can oxidize to form ferric hydroxide $[\text{Fe}(\text{OH})_3]$, which can further dehydrate to form iron oxide (Fe_2O_3) which is a major component of rust. This is shown in equation (9)



The pitting mechanism is as a result of chloride (Cl^-) ions accumulating at certain sites on the cast iron surface breaking down the protective film. This leads to localized anodic dissolution and the formation of pits which is a localized corrosion called pitting corrosion.

Linear Polarization Resistance data of the samples

The results of Linear Polarization Resistance of the samples used in this research are presented in table 4.2 and figure 4.3.

Sample	NH_3 (cm^3)	Corrosion current (A)	Current density (A/cm^2)	Potential (V)	R_p ($\Omega \cdot \text{cm}^2$)	B_c (V/dec)	B_a (V/dec)
Control 1	0	2.74E-05	2.16E-05	-0.772	938.10	-9.382	9.161
A1	5	1.73E-05	1.36E-05	-0.818	1488.00	-8.518	1.771
B1	10	1.83E-05	1.44E-05	-0.898	1401.00	-11.190	6.224
C1	15	2.44E-05	1.92E-05	-0.871	1054.00	-10.590	8.812
D1	20	3.03E-05	2.39E-05	-0.910	848.30	-9.379	9.434
E1	25	3.61E-05	2.84E-05	-0.932	711.50	-11.800	8.346

Table 4.2 Table for the Linear Polarization Resistance corrosion rate measurement technique

From table 4.2, it can be seen that the polarization resistance increased when the concentration of the aqueous ammonia was between (5- 15 cm^3). It could also be seen that the polarization

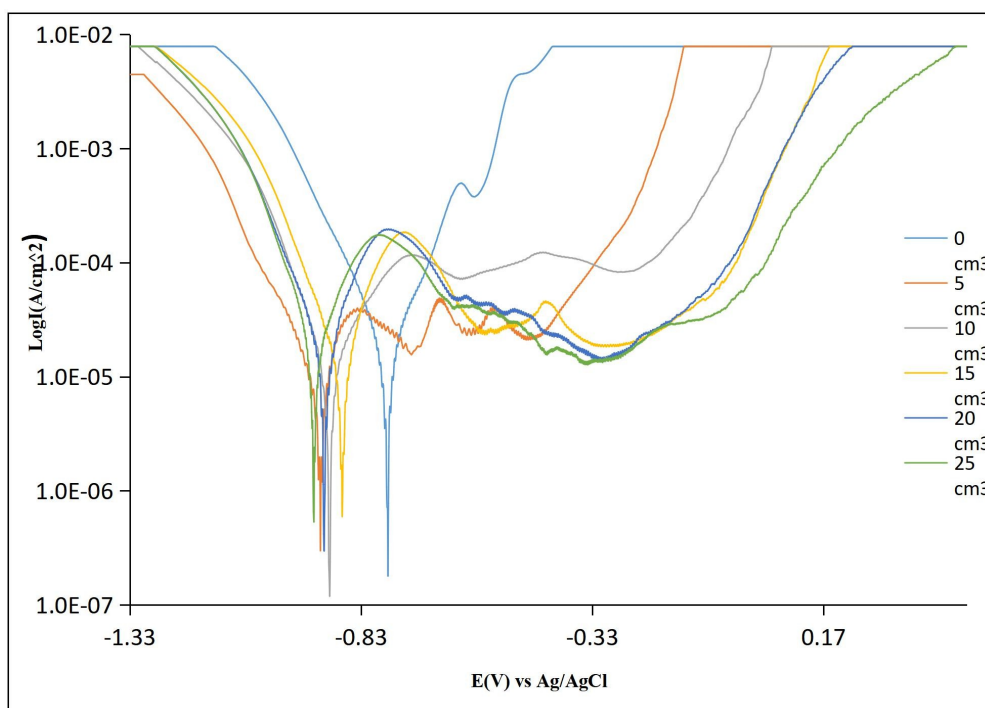


Figure 4.3: Graph for Linear Polarization Resistance corrosion rate measurement technique

resistance reduced when the concentration of the aqueous ammonia was increased between (20-25cm³) because of increase in corrosion rate.

The results also show from table 4.2, that the polarization resistance is the highest in sample A1 with a polarization resistance of 1488Ω·cm². This means the most effective and suitable concentration of inhibitor identified in this study is that of sample A1. The polarization resistance was least in sample E1 with a polarization resistance of 711.5Ω·cm². This means sample E1 is the least effective condition.

From table 4.2, the corrosion potential can be seen to increase from sample A1 to E1 negatively.

This indicates the increase in the tendency in corrosion as the corrosion potential increases negatively.

Figure 4.3 is also a graph that shows the relationship between log current density and corrosion potential (V) for samples control 1, A1, B1, C1, D1, E1. It can be seen that for the samples corrosion rate is directly proportional to the corrosion current density. Sample A1 possessed the lowest corrosion current density hence while has the lowest corrosion rate and sample E1 had the highest corrosion density hence why it had the highest corrosion rate. From figure 4.3, it can also be seen that there is a shift in the position of the anodic and cathodic tafel curves to lower current densities resulted to lower corrosion rates in the samples.

The corrosion potential is inversely proportional to polarization resistance. Hence as corrosion potential increased the polarization resistance reduced which also increases corrosion rate in the samples. (Abechi et al., 2020).

Inhibition efficiency data of the samples

The inhibition-efficiency results of the samples used in this research is presented in table 4.3.

Sample ID	Aqueous ammonia	
	concentration (cm ³)	Inhibition efficiency(%)
Control 1	0	0
A1	5	37.05
B1	10	33.11
C1	15	11.08
D1	20	0
E1	25	0

Table 4.3 Table for inhibition efficiencies of the samples

From table 4.3, it can be seen that the inhibition efficiency reduced when the concentration of the aqueous ammonia increased, which led to the increase of the concentration of ammonium chloride which increases the rate of corrosion just as Sun, A. and Fan, D., (2010) reported.

The results also show from table 4.3, that the highest inhibition efficiency is in sample A1 (37.05%) when compared to the uninhibited control. This means the most effective and suitable concentration of inhibitor identified in this study is that of sample A1. The inhibition efficiency is the least in sample E1 (0%) when compared to uninhibited control. This means sample E1 is the least effective condition.

The inhibition efficiency reduced because of the corrosion rate increased as a result of increase in the concentration ammonium chloride in the samples (Sun, A. and Fan, D., 2010).

Results for Gibbs Free Energy

The Gibbs free-energy results of the samples used in this research is presented in table 4.4

Sample ID	NH ₃			Equilibrium	
	Inhibition Efficiency (%)	NH ₃ Concentration (Molarity)	Surface Coverage (θ)	Constant of adsorption (K)	Gibbs Free Energy, ΔG (kJ mol ⁻¹)
Control 1	0	0	0	0	0
A1	37.05	2.94	0.370	0.2	-5.97
B1	33.11	5.87	0.331	0.1	-3.82
C1	11.08	8.81	0.111	0.0	0.60
D1	-10.48	11.74	-0.105	0.0	-
E1	-31.71	14.68	-0.317	0.0	-

Table 4.4 Table for Gibbs Free Energy results

From table 4.4, It can be seen that the values of ΔG moves from negative to positive from samples A1 to C1 which indicates that adsorption of the inhibitor on the metal surface changes from being spontaneous and feasible to non-spontaneous and non-feasible. Samples D1 and E1

do not possess value of ΔG which indicates there was no adsorption (Li et al., 2017; Akintubulumo et al., 2020).

From table 4.4, it can also be seen that the highest negative value for the Gibbs free energy is in sample A1 with a value of -5.97kJ/mol, this makes it the most effective and suitable concentration of inhibitor identified in this study. Samples D1 and E1 do not possess any values for Gibbs free energy which makes them the least effective conditions.

The highest negative value of Gibbs free energy can be seen in sample A1 which indicates that adsorption of the inhibitor on the metal surface is spontaneous, feasible and occurred according to the mechanism of physical adsorption. While samples D1 and E1 do not have any values for Gibbs free energy which indicates there was no adsorption (Li et al., 2017; Akintubulumo et al., 2020).

Results for Frumkin adsorption isotherm

The result of the Frumkin adsorption isotherm of the samples used in this research is presented in table 4.5 below

Sample ID	NH3 Surface Coverage		
	(θ)	$\theta/1-\theta$	α
A1	0.370	1.73	0.07
B1	0.331	2.91	-2.76
C1	0.111	1.10	-82.68
D1	-0.105	-1.11	0
E1	-0.317	-3.53	0

Table 4.5 Frumkin adsorption isotherm

From table 4.5, it can be seen the values of α decreased from sample A1 to sample E1. From table 4.5, it can also be seen that value for α for sample A1 is 0.07 which makes it the most effective and suitable concentration of inhibitor identified in this study.

The values of α for samples D1 and E1 are too negative and are represented as zero which makes them the least effective conditions. For sample A1, there was an attractive behavior of the aqueous ammonia on the surface of the cast iron. When concentration of aqueous ammonia was further increased for samples D1 and E1, the aqueous ammonia repulsed on the surface of the cast iron which led to the increase in the corrosion rate for the samples (Ituen et al., 2017; Gusti et al., 2023).

Results for Langmuir adsorption isotherm

The result of the Langmuir adsorption isotherm of the samples used in this research is presented in table 4.6 below.

Sample ID	NH3	MTH	K
	Concentration (Molarity*103)	Concentration/ θ	
A1	2.9E+00	7.9E+00	0.19
B1	5.9E+00	1.8E+01	0.08
C1	8.8E+00	8.0E+01	0.01
D1	1.2E+01	-1.1E+02	0
E1	1.5E+01	-4.6E+01	0

Table 4.6 Langmuir adsorption isotherm

From table 4.6, it can be seen that the adsorption value K reduced from sample A1 to sample E1 as the aqueous ammonia concentration was increased.

From table 4.6, it can also be seen that value for K for sample A1 is 0.2 which makes it the most effective and suitable concentration of inhibitor identified in this study. The values of K for samples D1 and E1 are zero which makes them the least effective conditions.

The highest value of K (0.2) can be seen for sample A1 which indicates higher adsorption strength compared to the other samples and the lowest values of K for samples D1 and E1 (0) which indicate no adsorption strength (Ituen et al., 2017; Gusti et al., 2023).

Result for Freundlich adsorption isotherm

The result of the Freundlich adsorption isotherm of the samples used in this research is presented in table 4.7.

Sample ID	NH3	Aqueous ammonia	
	Surface Coverage (θ)	Concentration (cm ³)	1
A1	0.37	5	0.57
B1	0.33	10	0.68
C1	0.11	15	-1.01
D1	-0.10	20	0
E1	-0.32	25	0

Table 4.7 Freundlich adsorption isotherm

From table 4.7, it can be seen that the constant ¹ increased from sample A1 to sample C1 as the aqueous ammonia concentration was increased and when the aqueous ammonia concentration

was further increased, the constant ¹ reduced to zero.

From table 4.7, it can also be seen that the values for the constant 1 is between 0 and 1 for sample

A1 (0.58) and sample B1(0.68) which makes them the most effective and suitable concentration of inhibitor identified in this study and the values for the constant 1 for samples D1 and E1

are too negative and are represented as zero which makes them the least effective conditions.

For samples A1 and B1 the adsorption is easy which resulted to the value of the constant 1 being between 0 and 1 and for samples D1 and E1 there was adsorption since their values of 1 is zero

(Ituen et al., 2017; Gusti et al., 2023).

4.3 Microstructure Results

The results of the SEM/EDS experiment of the samples used in this research are presented in figure 4.4 to figure 4.9 below.

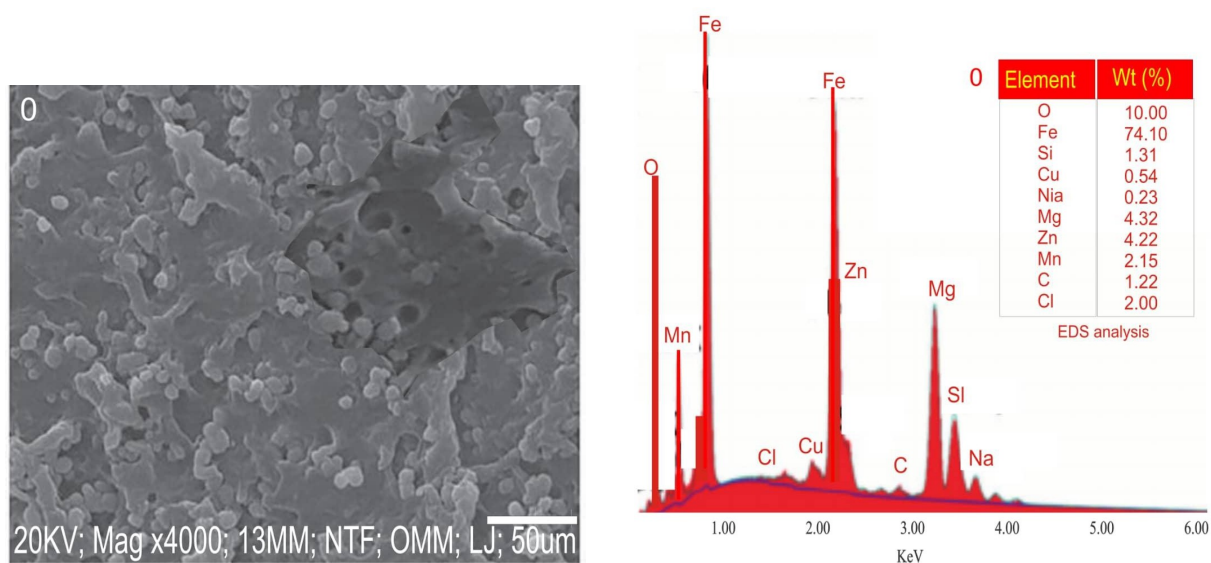


Figure 4.4: SEM image and EDS spectra for Sample Control 1

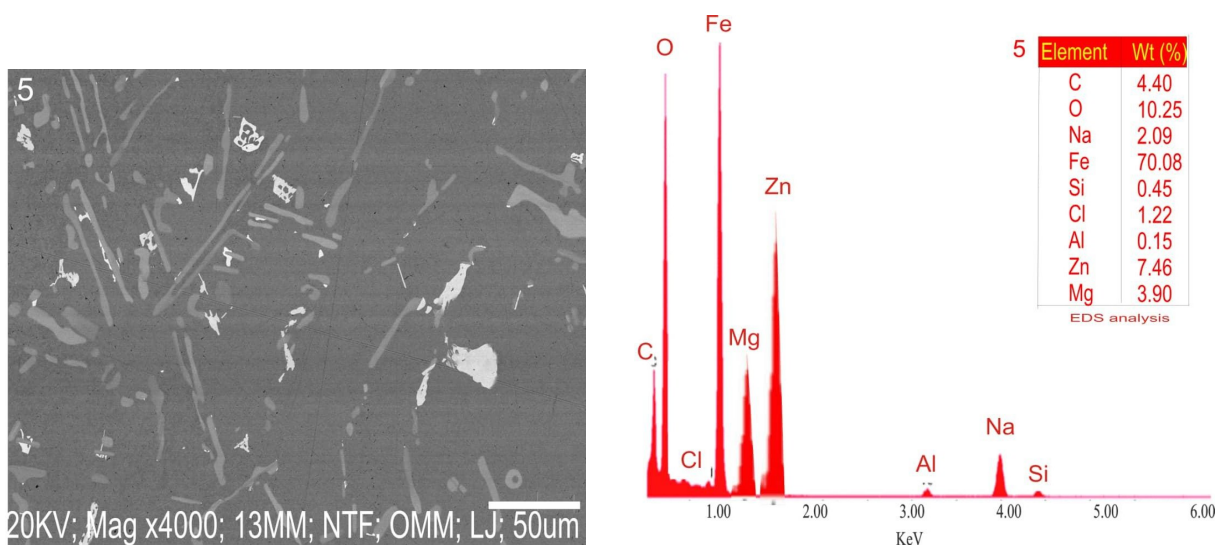


Figure 4.5: SEM image and EDS spectra for Sample A1

It can be seen from the figures above that the number of pitting corrosion decreased when the inhibitor concentration was between (5-15cm³) when compared to the uninhibited control

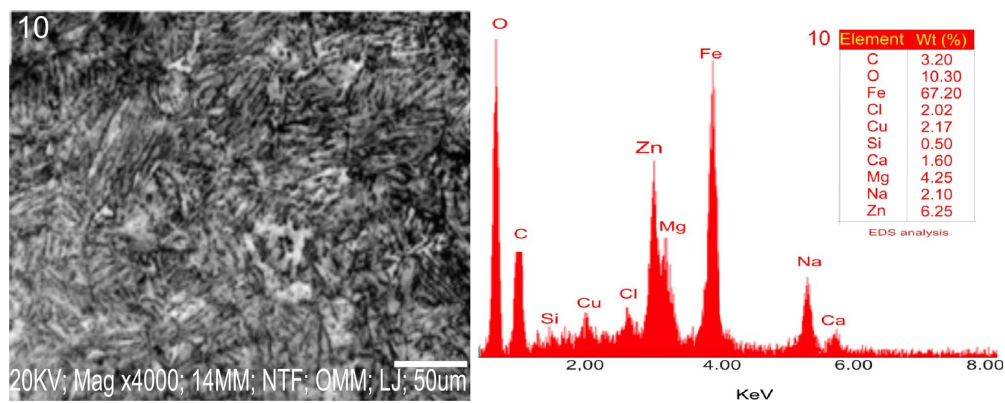


Figure 4.6: SEM image and EDS spectra for Sample B1

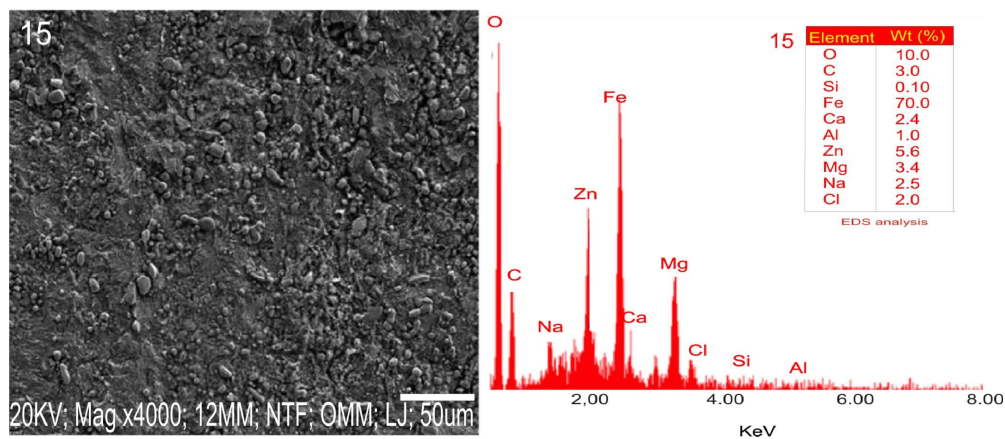


Figure 4.7: SEM image and EDS spectra for Sample C1

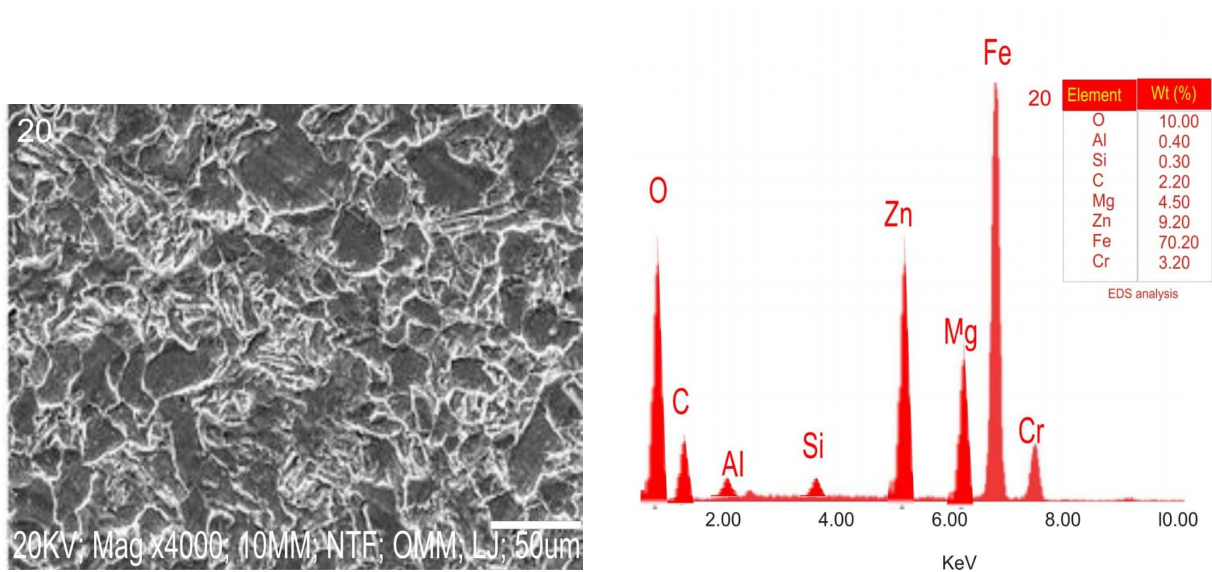


Figure 4.8: SEM image and EDS spectra for Sample D1

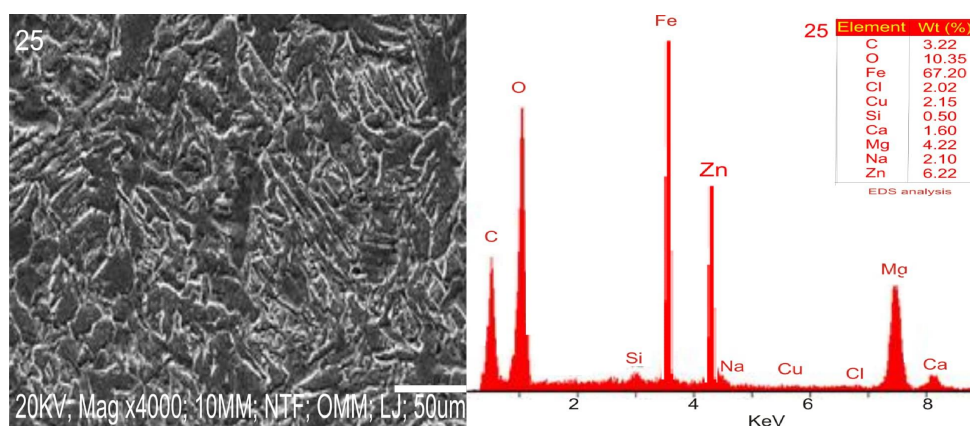


Figure 4.9: SEM image and EDS spectra for Sample E1

and the number of pitting corrosion increased when the inhibitor concentration was between (20-25cm³) It can also be seen from the figures above, that the number of pitting corrosion spots is least in sample A1 when compared to the uninhibited control, this makes sample A1 the most effective and suitable concentration of inhibitor identified in this study. The most number of pitting corrosion spots can be seen in samples E1 which makes it the least effective condition.

From the EDS spectra in the figures above, it can be seen that the concentration of both oxygen and chlorine is the least in sample A1 when compared to the uninhibited control which results for the low number of pitting corrosion spots and the concentration of both oxygen and chlorine is the highest in sample E1 when compared to control 1 which is the result for high number of pitting corrosion spots (Alotaibi et al., 2023).

5. Conclusion

The inhibitor concentration was successfully formulated and corrosion studies were successfully carried out, detailed structural morphologies were also carried out. Samples A1, B1 and C1 had corrosion rates which were less than the uninhibited control which was the uninhibited sample. The sample with the lowest corrosion rate was sample A1 with a corrosion rate of 0.166mm/year and with an inhibition efficiency of 37.05%.

The corrosion rates of samples D1 and E1 were discovered to be greater than that of the control sample 1, samples D1 and E1 had corrosion rates of 0.292mm/year and 0.348mm/year respectively and inhibition efficiencies of 0%.

The SEM images show fewer pitting corrosion spots for samples A1, B1 and C1 when compared to the uninhibited control. The SEM images also show more pitting corrosion spots for samples D1 and E1 when compared to the uninhibited control. Since the highest inhibition efficiency (37.05%) is less than 50%, the inhibitor cannot be commercially applied.

From this research, it can be concluded that human urine can fairly be used as an organic corrosion inhibitor for cast iron sodium chloride solution.

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